Backtesting Probability of Default Models

On the Statistical Validity of Multi-Period Backtesting Methods

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Internal Ratings-Based Modeling Process

- The development of an Internal Ratings-Based (IRB) model consists of two main areas: the development of the risk differentiation function and risk quantification.
- Development of the risk differentiation function is the first step in the overall model development process. The main goal of this step is to ensure the model can adequately rank risk, providing meaningful differentiation between low- and high-risk obligors or facilities.
- After developing the risk differentiation function, practitioners perform the risk quantification process, which essentially involves assigning an appropriate risk level to the output of the risk differentiation function.
- Risk quantification typically consists of specific steps such as: assignment of the base or best estimate, quantification of appropriate adjustments (AA), quantification of the margin of conservatism (MoC), and quantification of any other additional conservatism, for instance that required by certain local regulators. The first two steps are typically referred to as calibration.
- These steps apply to all three risk parameters: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD).

Probability of Default Backtesting

- In practice, backtesting is one of the most comprehensive validation exercises, typically investigating model estimates across different portfolio subsegments and time dimensions.
- Backtesting refers to assessing the appropriateness of model estimates, with the primary purpose of ensuring that they accurately predict the occurrence of defaults, i.e., that PD estimates provide reliable forecasts of default rates.
- One commonly applied approach is to backtest estimates over a recent period of three to five years, employing the z-score test.
- The z-score test is traditionally used in both one-sided and two-sided versions, often to assess evidence of underestimation or the accuracy (sometimes referred to as precision) of the PD estimate compared to the observed default rate.
- As necessary inputs, the z-score test requires the observed default rate, the calibrated or tested PD, and the sample size or number of exposures used for the test.
- Given that the calibrated PD is typically defined as the arithmetic average of yearly default rates, the question arises of how to properly construct multi-period inputs (the observed default rate and sample size) to align with the method used to calibrate the PD.

Probability of Default Backtesting cont.

- In practice, this multi-period average testing is typically performed by preparing the inputs - the observed default rate and the number of exposures - either by:
  1. calculating the observed default rate as a default-weighted average, with the total number of exposures equal to the sum of exposures across individual years;
  2. calculating the observed default rate as an arithmetic average of yearly default rates, with the number of exposures calculated in the same way.
- Neither of the above approaches aligns with the method used to calibrate the PD, which is based on the arithmetic average of yearly default rates.
- To address this shortcoming and the lack of statistical soundness in the PD backtesting exercise, Andrija Djurovic proposes a method that incorporates the specificities of the calibrated PD and can be easily extended to account for potential correlation structures between time periods.
- The purpose of this presentation is to provide a brief overview of the multi-period estimator's mean and variance that align with the calibrated PD approach, and to compare the statistical validity of this estimator with those commonly used in practice.

Multi-Period Estimator

- Running the z-score test for the purpose of PD backtesting requires the observed default rate, the calibrated or tested PD, and the sample size or number of observations. However, since the calibrated PD is provided by the model and calculated as an arithmetic average of yearly default rates, the other inputs need to align with this approach to ensure a statistically valid hypothesis testing framework.
- Both the average observed default rate and the effective sample size need to be aligned with the variance of the estimator.
- The multi-period estimator has the general form $\frac{1}{T}\sum_{i=1}^{T} ODR_i$, where ODR_i is the observed default rate at reference date i, and T is the number of reference dates.
- It can be shown that the variance of this estimator is $\frac{\overline{ODR}\cdot(1-\overline{ODR})}{T^2} \cdot \frac{1}{\sum_{i=1}^{T} \frac{1}{n_i}}$.

Multi-Period Estimator cont.

- Under the assumption that the average observed default rate follows a binomial proportion, its variance is given by $\frac{\overline{ODR}(1-\overline{ODR})}{N_e}$, where N_e is the effective sample size.
- By equating the variance of the average default rate with the variance of the estimator, practitioners obtain the effective sample size as $\frac{T^2}{\sum_{i=1}^{T} \frac{1}{n_i}}$.
- Practitioners should note that the above formula for the effective sample size assumes that default rates over time are independent, while defaults within each period follow a binomial distribution. This approach can be extended to incorporate different correlation structures of default rates over time, which is particularly important when overlapping periods are used for calculating default rate.
- To verify whether this testing framework is correctly specified, this presentation includes a simulation study that investigates the distribution of p-values under the true null hypothesis and presents the size of the Type I error for all three methods discussed.

Simulation Study

The purpose of this simulation is to compare the distribution of p-values under the true null hypothesis for three methods used in multi-period backtesting of the average default rate.

The simulation is based on the following inputs:

1. The number of exposures per reference date: 143, 122, 212, 190, and 361.
2. The average default rate is set equal to the calibrated PD at 5%.

The tested methods are labeled as follows:

1. *Method 1*: the average default rate (ODR) is calculated as the arithmetic mean of reference date default rates, with the number of exposures calculated as the effective sample size (N) for the estimator of the average default rate. This is the method proposed by Andrija Djurovic.
2. *Method 2*: the average default rate (ODR) is calculated as a default-weighted mean, with the number of exposures (N) equal to the sum of exposures across reference dates.
3. *Method 3*: both the average default rate (ODR) and the number of exposures (N) are calculated as the arithmetic mean of reference date default rates and the number of exposures, respectively.

Simulation Study cont.

Given the inputs and the description of the tested methods from the previous slides, the following steps outline the main simulation setup:

1. Select the number of reference dates (`T`) for testing. In this case, it ranges from 2 to 5, with the number of exposures chosen according to the given order in the previous slide. For example, if `T` = 3, then the number of exposures per reference date is: 143, 122, 212. The same logic applies to any other `T` in the range from 2 to 5.
2. For each reference date, simulate the number of defaults from a binomial distribution with a success probability equal to `PD` (5%).
3. Based on the simulated defaults from step 2 and the number of exposures from step 1, recalculate the simulated default rate.
4. Run the z-score test by comparing the average default rate based on the simulated values from step 3 (under the three different methods) with the calibrated `PD` (5%), and store the resulting `p-value`.
5. Repeat steps 2 to 4 a total of `100,000` times and record the `p-values`.
6. Calculate the Type I error as the proportion of simulated `p-values` below the 5% significance level.

Given the above setup, for the method that correctly specifies the testing framework, the expectation is that the simulated `p-values` are approximately uniformly distributed and that the Type I error is approximately equal to the selected significance level of 5%.

The following slides present the basics of the z-score test used in this simulation, as well as the testing results.

Simulation Study cont.

The Z-Score Test

The test statistic of the z-score test is given as follows:

$$z_{score} = \frac{ODR - PD}{\sqrt{\frac{PD(1-PD)}{N}}}$$

where:

- ODR is the observed default rate;
- PD is the calibrated PD;
- N is the number of exposures.

Under the assumption that the z_{score} test statistic follows a standard normal distribution, a `p-value` is calculated accordingly.

Simulation Results

```
## T    pd   method type_I_error
## 2 5.00% Method 1      5.0090%
## 2 5.00% Method 2      4.5780%
## 2 5.00% Method 3      0.6120%
## 3 5.00% Method 1      5.0720%
## 3 5.00% Method 2      4.5940%
## 3 5.00% Method 3      0.1410%
## 4 5.00% Method 1      4.9730%
## 4 5.00% Method 2      5.0400%
## 4 5.00% Method 3      0.0110%
## 5 5.00% Method 1      4.9410%
## 5 5.00% Method 2      4.4870%
## 5 5.00% Method 3      0.0070%
```

Simulation Results cont.